Project Roadmap

Project Title

Tile Survival Game

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1. Project Description

Tile Survival is a multiplayer survival platform game developed using **Python and Pygame**.

Players control characters that can:

- Move **left and right**
- Move **up and down across tiles**
- **Jump between platform tiles**

The game world consists of a platform made of **square tiles**. These tiles **randomly disappear after warning animations**, forcing players to move strategically to avoid falling.

The objective is simple:

Stay on the platform as long as possible.

As the game progresses:

- tiles disappear faster
- hazards such as bullets and traps appear
- the difficulty gradually increases

The game will support:

- Single Player Survival
- Local Multiplayer
- Player vs Computer
- Online Multiplayer (LAN / Internet)

In addition to the game, the project includes:

- A **deployment website**
- A **player analytics dashboard**

- **Social media promotion**

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2. Project Objectives

The project aims to:

1. Develop a **2D multiplayer survival platform game**.
2. Implement a **dynamic tile disappearance system**.
3. Implement **AI opponents**.
4. Implement **network multiplayer gameplay**.
5. Deploy the game through a **dedicated website**.
6. Collect **player analytics**.
7. Promote the game through **social media marketing**.

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3. Core Game Features

Player Mechanics

Players can:

- Move left and right
- Move across tiles
- Jump between tiles
- Avoid disappearing tiles
- Avoid hazards

Tile System

Tiles will have **three states**.

State	Description
Normal	Tile is stable
Warning	Tile flashes before disappearing

State	Description
Disappeared	Tile no longer exists

This creates dynamic gameplay where players must constantly move.

Difficulty Scaling

As the game progresses:

- Tile disappearance speed increases
- Multiple tiles disappear simultaneously
- Hazards appear

Examples of hazards:

- bullets
- moving traps
- enemy AI

Multiplayer Modes

The game will support:

1. Single Player Mode
2. Local Multiplayer Mode
3. Player vs Computer
4. LAN Multiplayer
5. Online Multiplayer

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4. Technology Stack

Game Development

- Python
- Pygame

Networking

- Python Socket Programming

Website

- React
- Node.js
- Express.js

Analytics

- Google Analytics

Metrics collected:

- number of players
- session duration
- downloads
- player retention

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5. Team Collaboration Structure (8 Members)

Instead of separating the project into different departments, **all 8 members work together each day toward the same milestone**.

Each member contributes to:

- coding
- testing
- debugging
- design
- documentation

Tasks are distributed dynamically depending on the needs of the day.

This structure encourages:

- collaboration
- faster debugging
- knowledge sharing

- better integration of features

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6. Development Schedule (Daily Goals)

The development cycle runs in **weekly sprints**.

Every **Tuesday**, the team presents a **playable version of the game** showing the progress made during the week.

Week 1 — Core Game Prototype

Wednesday — Project Setup

Goals:

- Setup GitHub repository
- Setup Pygame environment
- Create base project structure
- Create initial game window
- Setup basic game loop

Thursday — Player Movement

Goals:

- Implement player character
- Implement movement:
- move left
- move right
- basic animations

Friday — Tile Grid System

Goals:

- Implement tile grid platform

- Render square tiles
- Player collision with tiles

Saturday — Tile Warning System

Goals:

- Implement tile state system:
- normal
- warning
- disappeared
- Add warning animation

Sunday — Tile Disappearance Logic

Goals:

- Random tile disappearance
- Player falls when tile disappears
- Add respawn or elimination logic

Monday — Game Polishing

Goals:

- Improve player movement
- Fix bugs
- Improve tile animations
- Ensure basic gameplay works smoothly

Tuesday — Presentation 1

Deliverable:

A **playable prototype** where:

- player moves
- tiles disappear
- player can fall

Week 2 — Gameplay Expansion

Wednesday

Goals:

- Implement **jump mechanics**
- Improve physics

Thursday

Goals:

- Add **multiple players**
- Implement local multiplayer controls

Friday

Goals:

- Add **hazard system**
- Introduce bullets

Saturday

Goals:

- Add **moving traps**

Sunday

Goals:

- Implement **difficulty scaling**

Monday

Goals:

- Gameplay balancing
- Fix bugs
- Improve visual feedback

Tuesday — Presentation 2

Deliverable:

Playable game with:

- jumping
- hazards
- difficulty progression
- local multiplayer

Week 3 — AI and Multiplayer

Wednesday

Goals:

- Create **basic AI player**

Thursday

Goals:

- Improve AI movement
- AI avoids disappearing tiles

Friday

Goals:

- Implement **LAN multiplayer**

Saturday

Goals:

- Player synchronization

Sunday

Goals:

- Multiplayer testing

Monday

Goals:

- Fix networking bugs
- Optimize gameplay

Tuesday — Presentation 3

Deliverable:

Playable game with:

- AI opponents
- LAN multiplayer

Week 4 — Website and Deployment

Wednesday

Goals:

- Create deployment website

Thursday

Goals:

- Implement **download page**

Friday

Goals:

- Create **leaderboard system**

Saturday

Goals:

- Integrate **analytics**

Sunday

Goals:

- Connect game data to website

Monday

Goals:

- Final testing
- Bug fixing
- Game polishing

Tuesday — Final Presentation

Deliverables:

1. Fully playable **Tile Survival Game**
2. Multiplayer system
3. Deployment website
4. Analytics dashboard

5. Social media promotion campaign

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7. Expected Deliverables

By the end of the project, the team will produce:

1. A fully playable **2D survival game**
2. A **multiplayer gameplay system**
3. A **deployment website**
4. A **player analytics dashboard**
5. A **social media marketing campaign**

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8. Future Improvements

Future versions may include:

- global online matchmaking
- character skins
- ranked competitive mode
- seasonal events
- mobile version